

Homework 4

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Problem 1. Given an example of an open subset $U \subseteq \mathbb{C}$ and two holomorphic functions $f, g : U \rightarrow \mathbb{C}$ such that $f' = g'$ but g does not equal f plus a constant.

Proof. Let U be the union of two disjoint open connected subsets $U_1, U_2 \subseteq \mathbb{C}$. Define $f(z) = z^2$. Define $g|_{U_1} = z^2$ and $g|_{U_2} = z^2 + 1$. Clearly f is holomorphic. Since $g|_{U_1}, g|_{U_2}$ are holomorphic, since $U_1 \cup U_2 \cong U_1 \amalg U_2$, and since coproducts of holomorphic functions are holomorphic, it follows that g is holomorphic. Moreover, $\partial_z f = \partial_z g = 2z$. However, they do not differ by a constant, but rather by the function

$$z \mapsto \begin{cases} 0, & z \in U_1 \\ 1, & z \in U_2 \end{cases}$$

which is only locally constant. Since U is disconnected, this does not imply it is constant. Since disjoint connected components exist in $\mathbb{C}$ (consider two disjoint open balls), this is an example. ■

Problem 2. Let $f(z) = z^2$. Calculate that the integral of f around the circle $\partial B_1(2)$ given by

$$\int_0^{2\pi} f(2 + e^{i\theta}) d\theta$$

is not zero. Yet the Cauchy theorem asserts that

$$\oint_{\partial B_1(2)} f = 0.$$

Give an explanation.

Proof. Although $2 + e^{i\theta}$ parameterizes $\partial B_1(2)$ counterclockwise for $\theta \in [0, 2\pi]$, the line integral of f along γ is defined as

$$\int_{\gamma} f dz = \int_0^{2\pi} \gamma^*(f dz) = \int_0^{2\pi} f(\gamma(\theta)) \gamma'(\theta) d\theta.$$

Note that the given integral is missing the factor of γ' . We compute the given integral:

$$\int_0^{2\pi} 4 + 4e^{i\theta} + e^{2i\theta} d\theta = 8\pi$$

since $e^0 = e^{i2\pi} = e^{i4\pi}$. ■

Problem 3. 1. Let γ be a positively oriented parameterization of S^1 with a counterclockwise orientation. Give an example of a C^1 function f on a neighborhood of $\overline{B_1(0)}$ such that

$$\int_{\gamma} f = 0$$

but such that f is not holomorphic on any open set.

2. Suppose that f is C^0 on $B_1(0)$ and satisfies

$$\int_{\partial B_r(0)} f = 0$$

for all $0 < r < 1$. Must f be holomorphic on $B_1(0)$?

Proof. (1) Let $f(z) = \bar{z}^2$, and let $z(\theta) = e^{i\theta}$ such that $dz = ie^{i\theta}d\theta$. Then

$$\int_{\gamma} f = \int_0^{2\pi} e^{-2i\theta} i e^{i\theta} d\theta = - \int_0^{2\pi} e^{-i\theta} d\theta = 0.$$

However, clearly $\partial_{\bar{z}}f = 2\bar{z} \neq 0$ on all open $U \subseteq \mathbb{C}$.

(2) Take f the same as in (1), and parameterize $\partial B_r(0)$ by $z(\theta) = re^{i\theta}$. The integral is then

$$\int_0^{2\pi} rie^{-i\theta} d\theta,$$

which is still 0. ■

Problem 4. Give a simple proof (actually a one-line proof if you use what we know about power series) that an analytic function has a local primitive.

Proof. Any power series clearly has a primitive, and analytic functions admit local power series representations. ■